MIL-T-60677(MU)
7 October 1968

SUPERSEDING
FA-PD-MI-2664
11 June 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M9
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated: M9, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10520479	- Thruster, Cartridge Actuated: M9, Metal Parts Assembly
C8845196	- Container, Ammunition, Fiber M461 for Thruster, Cartridge Actuated: M9 with Cartridge M119

FSC 1377

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C8845197 - Box, Packing, Ammunition for Thruster; Cartridge Actuated, M9, with Cartridge M119 in Fiber Container M461
IEL-10520479 - Inspection Equipment List for Thruster, Cartridge Actuated; M9, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Applications for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.- Components for remanufacture shall be obtained from the M9 model thruster. All components from earlier model types shall be modified in accordance with the latest component drawing. All components from earlier model types shall be identified and segregated for required modification prior to reassembly (See 6.4.3).

3.3 Hydrostatic pressure.- The body shall withstand a constant internal pressure of 10,000 pounds per square inch (psi), minimum, for 15 seconds, minimum without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.2).

3.4 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing, F10520479. All threaded joints shall be assembled with sufficient torque to lightly bottom components, except as noted in Table I.

3.4.1 Metal parts assembly shall be assembled with the following new metal part components: Firing Pin, Shear Pin, 10520486; Piston Shear Pins, 10520487; and O-rings, MS29513.

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3.4.2 Metal parts assembly shall be assembled with the following metal part components which may be modified to meet requirements of applicable drawing: Ring, C10520489; Adapter, C10520480; Body, F10520482; and Set Screws, Commercial #6-32 UNC-3Ax 1/8. Set screws or any other thread component which shows evidence of cross threading, stripping or excessive wear shall be replaced.

3.4.3 Ring rotation.- Upon completion of piston and adapter assembly as specified in Table I, the ring shall be free to rotate 360 degrees.

3.4.4 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

Components	A Seating or Bottoming (inch pounds)	B Breakaway (inch pounds)
Piston and Adapter	100 ± 20	75 min

3.4.5 Trunnion rotation.- Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings, and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish re-applied (see 6.4).

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when apssed in front of the compass at a distance of 6 inches (see 6.3).

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following·

3.7.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or any other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 and as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Products" provisions of MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lots of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be either reprocessed or rejected.

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4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample sizes for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.2.2.3 Adapter, Drawing C10520480 covering a detail of Drawing C10520481.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Body, Drawing F10520482 covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corners	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Breech, Drawing D10520483 covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp edge	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Collar, Drawing C10520484 covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Pin, Firing, Drawing C10520485 covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp edges	Visual
104. Foreign matter	Visual

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.8 Piston, Drawing D10520488 covering a detail of Drawing F10520479.	
CRITICAL	
None defined	
MAJOR	
101. Length from head to thread undercut	Gage
102. Thread damaged or not full	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp edge	Visual
106. Foreign matter	Visual
MINOR	
None defined	
4.2.2.9 Ring, Drawing C10520489 covering a detail of Drawing F10520479.	
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.10 Trunnion, Drawing D10520490 covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter of lugs	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Adapter and Pellet Assembly, Drawing C10520481, covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pellet loose, missing or improperly inserted	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Breech and Firing Pin, Sub-Assembly, consists of: Breech 10520483, Firing Pin 10520485 and Packing MS29513-9, covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
1. Firing pin protrusion	Gage

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4.2.2.12 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Any part missing	Visual
102. Thread damaged or not full	Visual
103. Protective finish, inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Assembly, Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
*101. Ring not free to rotate 360 degrees	Manual
102. Trunnion not free to rotate 360 degrees	Manual
103. Any part missing, except as otherwise classified	Visual
104. Improper assembly	Visual
105. Thread damaged or not full	Visual
106. Thread compound on set screw missing	Visual
107. Set screws not filled with barrier material	Visual
108. Shipping plugs or shipping cap missing	Visual
109. Nylok pellet missing	Visual
110. Protective finish, inadequate or defective	Visual
111. Burr	Visual
112. Foreign matter	Visual

* Upon completion of Piston and Adapter assembly

MINOR	
None defined	

4.2.2.14 Container, Fiber, Drawing C8845196.

CRITICAL	
None defined	

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4.2.2.14 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.15 Box, Packing, Drawing C8845197.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each body will be subjected to the internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the component.

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4.2.3.2 Breakaway torque and curing time.- Each piston and adapter shall be subjected to a breakaway torque application after the thread sealant has been cured. Failure of any unit to comply with the requirements of 3.4.4, Table I and Drawing F10520479, as applicable, shall be cause for rejection of the unit.

4.2.3.3 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements, shall be cause for rejection of the lot.

4.2.3.4 Residual magnetism.- Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520479.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- The test shall be performed after the application of protective finish. The testing device used shall internally seal the body at each end without applying any axial load other than that applied by the liquid pressure.

4.3.2 Breakaway torque and curing time.- The piston shall be secured in a holding device and the torque shall be applied to the adapter. Curing shall be at room temperature for the length of time as specified on Drawing F10520479.

4.3.3 Protective finish.- Test procedure for protective finish shall be in accordance with the specifications referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass, in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Level A.- Packing and marking shall be in accordance with Drawings C8845196 and C8845197.

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5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.2 A mechanical failure.- Is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 When remanufacturing and loading of assembly are performed by the same facility, the requirement of 3.6 is waived and the residual magnetism test is performed on the loaded assembly only.

6.4 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.4.1 Protective finish.- Exterior components which include Collar, Trunnion, Body and Adapter shall be stripped of protective finish and replated in accordance with the applicable specifications and drawings.

6.4.2 Nylon pellets.- Components which contain pellets and are to be replated shall have new pellets installed after replating, in accordance with the applicable drawings.

6.4.3 The required modification shall be determined after components are identified and segregated

6.5 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2664 dated 11 June 1965.

Custodian:

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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